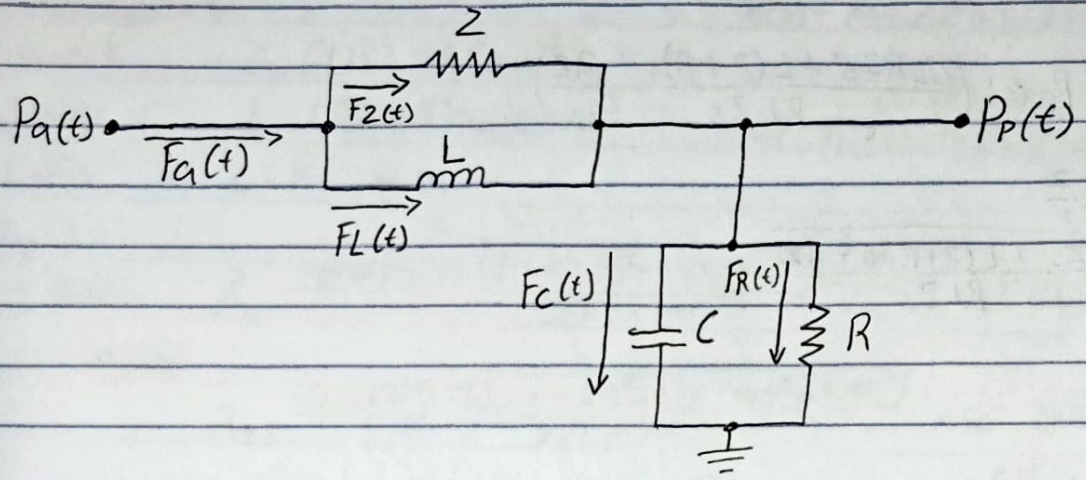
Modelado y Control del sistema cardiovascular mediante el modelo Windkessel
Practica 5.4:
05/oct/2026



* Función de transferencia: (ecuaciones Principales)

Pa(t) = Entrada ; Pp(t) = Salida

$F_a(t) = F_2(t) + F_L(t) = F_C(t) + F_R(t)$

$F_2(t) = \frac{P_a(t) - P_p(t)}{Z}$

$F_L(t) = \frac{1}{L}\int [P_a(t) - P_p(t)]\,dt$

$F_C(t) = C\frac{dP_p(t)}{dt}$

$F_R(t) = \frac{P_p(t)}{R}$

* Transformada de Laplace

$F_a(s) = F_2(s) + F_L(s) = F_C(s) + F_R(s)$

$F_2(s) = \frac{P_a(s) - P_p(s)}{Z}$

$F_L(s) = \frac{P_a(s) - P_p(s)}{Ls}$

$F_C(s) = Cs\,P_p(s)$

$F_R(s) = \frac{P_p(s)}{R}$

* Procedimiento algebraico:

$F_2(s) + F_L(s) = F_C(s) + F_R(s)$

$\frac{P_a(s) - P_p(s)}{Z} + \frac{P_a(s) - P_p(s)}{Ls} = Cs\,P_p(s) + \frac{P_p(s)}{R}$

$\frac{P_a(s)}{Z} - \frac{P_p(s)}{Z} + \frac{P_a(s)}{Ls} - \frac{P_p(s)}{Ls} = Cs\,P_p(s) + \frac{P_p(s)}{R}$

$P_a(s)\left(\frac{1}{Z} + \frac{1}{Ls}\right) = P_p(s)\left(Cs + \frac{1}{R} + \frac{1}{Ls} + \frac{1}{Z}\right)$

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$$P_a(s)\frac{Ls+Z}{LZs} = P_p(s)\left(\frac{CLRZs^2 + LZs + RZ + RLs}{RLZs}\right)$$

$$P_a(s)\frac{Ls+Z}{LZs} = P_p(s)\left(\frac{CLRZs^2 + L(Z+R)s + RZ}{RLZs}\right)$$

$$\frac{P_p(s)}{P_a(s)} = \frac{\frac{Ls+Z}{LZs}}{\frac{CLRZs^2 + L(Z+R)s + RZ}{RLZs}}$$

* Resultado:

$$\frac{P_p(s)}{P_a(s)} = \frac{LRs + RZ}{CLRZs^2 + L(Z+R)s + RZ}$$ } Función de transferencia.

* Estabilidad del sistema en lazo abierto

$$CLRZs^2 + L(Z+R)s + RZ = \emptyset$$

$a = CLRZ = 3.5\times10^{-5}$

$b = L(Z+R) = 1.01\times10^{-3}$

$c = RZ = 0.01$

$$\lambda_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-L(Z+R) \pm \sqrt{L^2(Z+R)^2 - 4CLR^2Z^2}}{2CLRZ}$$

$$\lambda_1 = \frac{-(1.01\times10^{-3}) + \left(\sqrt{(1.01\times10^{-3})^2 - 4(3.5\times10^{-5})(0.01)^1}\right)}{2(3.5\times10^{-5})} = -14.42857143 + 8.805146918\,i$$

Hipotenso

$Z = 0.010$

$C = 3.500$

$R = 1.000$

$L = 0.001$

$$\lambda_2 = \frac{-(1.01\times10^{-3}) - \left(\sqrt{(1.01\times10^{-3})^2 - 4(3.5\times10^{-5})(0.01)}\right)}{2(3.5\times10^{-5})} = -14.42857143 - 8.805146918\,i$$

$$\frac{P_p(s)}{P_a(s)} = \frac{LRs + RZ}{CLRZs^2 + L(Z+R)s + RZ} = \frac{(1\times10^{-3})s + 0.01}{(3.5\times10^{-5})s^2 + (1.01\times10^{-3})s + 0.01}$$

$\lambda_1 = -14.42857143 + 8.805146918\,i$

$\lambda_2 = -14.42857143 - 8.805146918\,i$

Por lo tanto, el sistema presenta una respuesta estable y subamortiguado a un escalón unitario de entrada.

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Normotenso

$$H(s) = \frac{LRs + RZ}{CLRZs^2 + L(Z+R) + RZ}$$

$Z = 0.010$ $\quad a = CLRZ = 1.5\times10^{-5}$

$C = 1.500$ $\quad b = L(Z+R) = 1.01\times10^{-3}$ $\quad H(s) = \dfrac{(1\times10^{-3})s + 0.01}{(1.5\times10^{-5})s^2 + (1.01\times10^{-3})s + 0.01}$

$R = 1.000$ $\quad c = RZ = 0.01$

$L = 0.001$

$\lambda_1 = -12.0616261$ $\quad \lambda_1 = \dfrac{-(1.01\times10^{-3}) + \left(\sqrt{(1.01\times10^{-3})^2 - 4(1.5\times10^{-5})(0.01)}\right)}{2(1.5\times10^{-5})} = -12.0616261$

$\lambda_2 = -55.27170723$

$$\lambda_2 = \frac{-(1.01\times10^{-3}) - \left(\sqrt{(1.01\times10^{-3})^2 - 4(1.5\times10^{-5})(0.01)}\right)}{2(1.5\times10^{-5})} = -55.27170723$$

Por lo tanto, el sistema presenta una respuesta estable y sobreamortiguada a un escalón unitario de entrada.

Hipertenso

$$H(s) = \frac{LRs + RZ}{CLRZs^2 + L(Z+R) + RZ} =$$

$Z = 0.010$ $\quad a = CLRZ = 1\times10^{-5}$

$C = 1.000$ $\quad b = L(Z+R) = 1.01\times10^{-3}$ $\quad H(s) = \dfrac{(1\times10^{-3})s + 0.01}{(1\times10^{-5})s^2 + (1.01\times10^{-3})s + 0.01} =$

$R = 1.000$ $\quad c = RZ = 0.01$

$L = 0.001$

$\lambda_1 =$ $\quad \lambda_1 = \dfrac{-(1.01\times10^{-3}) + \left(\sqrt{(1.01\times10^{-3})^2 - 4(1\times10^{-5})(0.01)}\right)}{2(1\times10^{-5})}$

$\lambda_2 =$

$\lambda_1 = -11.12678575$

$$\lambda_2 = \frac{-(1.01\times10^{-3}) - \left(\sqrt{(1.01\times10^{-3})^2 - 4(1\times10^{-5})(0.01)}\right)}{2(1\times10^{-5})}$$

$\lambda_2 = -89.87321425$

Por lo tanto, el sistema presenta una respuesta estable y sobreamortiguada a un escalón unitario de entrada.

* **Error en estado estacionario** (se calcula a un escalón unitario de entrada)

$$e(s) = \lim_{s\to\emptyset} s\, P_a(s)\left(1 - \frac{P_o(s)}{P_a(s)}\right) = e(s) = \lim_{s\to\emptyset} s\,\frac{1}{s}\left(1 - \frac{LRs + RZ}{CLRZs^2 + L(Z+R)s + RZ}\right) = \emptyset$$

$\therefore e(t) = \emptyset V.$

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* Modelo de ecuaciones integro-diferenciales

$$F_Z(t) + F_L(t) = F_C(t) + F_R(t)$$

$$\frac{P_a(t) - P_p(t)}{Z} + \frac{1}{L}\int [P_a(t) - P_p(t)]\,dt = C\frac{dP_p(t)}{dt} + \frac{P_p(t)}{R}$$

Nota: no se pueden despejar funciones que se están derivando o integrando.

$$\frac{P_a(t)}{Z} + \frac{1}{L}\int [P_a(t) - P_p(t)]\,dt - C\frac{dP_p(t)}{dt} = \frac{P_p(t)}{R} + \frac{P_p(t)}{Z}$$

$$\left(\frac{1}{R} + \frac{1}{Z}\right) P_p(t) = \frac{P_a(t)}{Z} + \frac{1}{L}\int [P_a(t) - P_p(t)]\,dt - C\frac{dP_p(t)}{dt}$$

$$P_p(t) = \left[\frac{P_a(t)}{Z} + \frac{1}{L}\int [P_a(t) - P_p(t)]\,dt - C\frac{dP_p(t)}{dt}\right]\frac{RZ}{Z+R} \quad \} \text{Modelo}$$

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